

(TR-104) PYTHON WITH AI/ML

Training Day 73 Report

Date: 22 April 2026

Topic Covered: FastAPI Backend Optimization, MongoDB Multi-Collection Integration & AI Agent Development

The seventy-third day of training focused on enhancing the MB Trailer FastAPI backend by implementing optimized MongoDB operations, caching mechanisms, intelligent query tools, and LangChain-based AI agent integration for efficient backend processing and analytics.

- Configured environment variables for MongoDB, Ollama, caching, and performance tuning using dotenv for centralized configuration management.
- Established asynchronous MongoDB connection using Motor with connection pooling support for better scalability and performance.
- Implemented multi-collection support including leasepayments, insurances, and trailers collections.
- Created dynamic index creation system across collections to optimize database query performance.
- Replaced deprecated startup events with FastAPI lifespan event handler for modern lifecycle management.
- Implemented in-memory caching system with TTL using time.monotonic to optimize repeated database queries.
- Developed utility functions for timezone-aware date handling and structured response formatting.
- Optimized repeated insurance aggregation logic using reusable MongoDB pipeline helper functions.
- Improved payment analytics tools using asyncio.gather for parallel execution and faster aggregation handling.
- Enhanced payment-related tools with filtering, sorting, upcoming payment tracking, and payment-method analysis features.
- Developed trailer-related tools for fetching trailer details using ObjectId, trailer number, and license plate.

- Created insurance-related tools for analytics, renewal tracking, provider-based analysis, and verification status monitoring.
- Implemented lookup-based MongoDB queries for combining trailer and insurance data efficiently.
- Integrated LangChain agent with Ollama model for structured tool-based AI query handling.
- Defined strict system prompts and mapped backend tools for controlled and accurate responses.
- Implemented asynchronous agent execution, response extraction logic, and robust exception handling inside FastAPI endpoints.
- Reviewed curriculum structure and analyzed learning flow and topic sequencing for better understanding of LangChain fundamentals.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/chatbot>

Overall, the training enhanced understanding of FastAPI optimization, asynchronous MongoDB operations, caching strategies, LangChain-based AI agents, intelligent tool integration, and scalable AI-powered backend system development.